Achieving 10% Revenue Growth: Inventory as the Key Execution Lever

Analysis on Product, Price, and Inventory

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Executive Summary

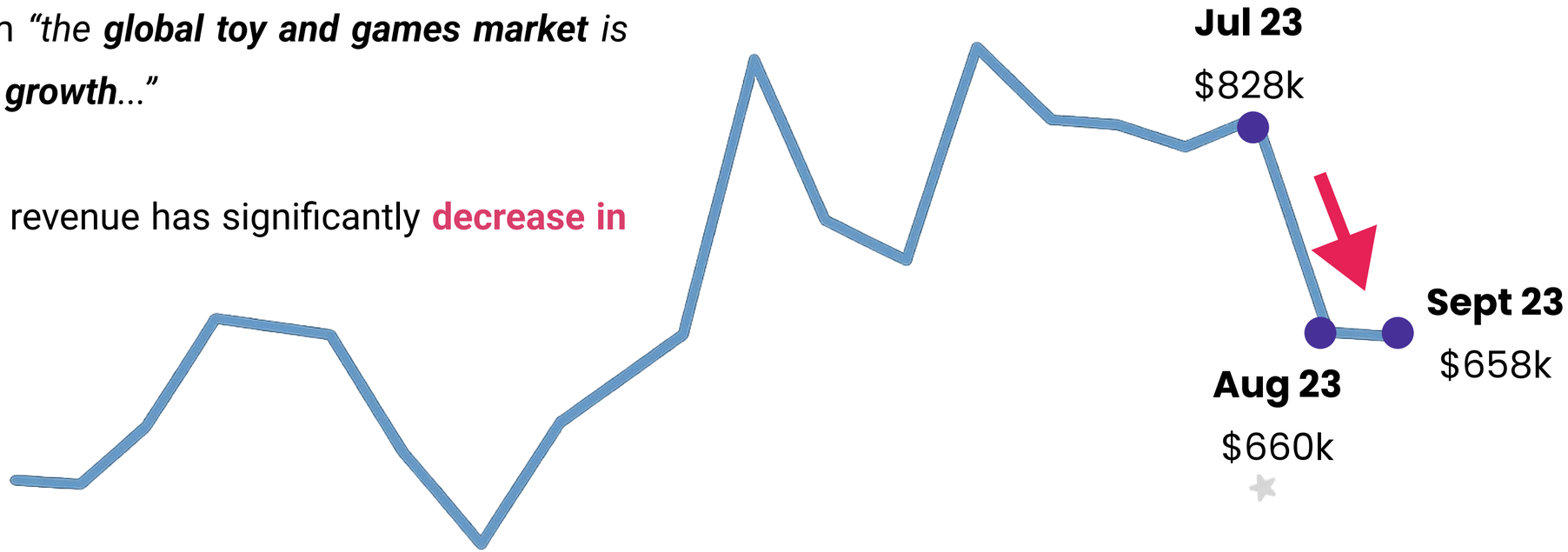
- **Project Description:** Analysis of a three-month continuous revenue decline to identify root causes and establish a 6-month recovery roadmap aimed at a 10% revenue increase.
- **Objective:** Identify key operational and market factors contributing to revenue leakage and underperformance.
- **Tools:** Python (Prophet for unit-demand forecasting) and Tableau (Macro-revenue and inventory visualization).
- **Metrics:** Total Revenue & Revenue Gap, Inventory Turnover, Weeks of Supply (WoS), Estimated Stockout Opportunity Revenue, Holiday Lift %
- **Analysis Result:**
  - **Revenue Leakage:** An estimated **$3.7M** is lost due to systemic stockouts.
  - **Inventory Imbalance:** 25% of products are at critical stockout risk, while the average inventory sits for **14 weeks**.
  - **Regional Failures: La Paz** and **Toluca** suffer from 100% stockout rates on core products, while **Ciudad de México** and **Morelia** are heavily overstocked in Games and Art & Crafts.
  - **Profit Leaders: Colorbuds** is the primary profit driver, while **Lego Bricks** drives high revenue through premium pricing.
  - **Forecast Risk:** While the overall trend is positive, a significant **dip is forecasted for September** despite local festivals.
- **Recommendation:**
  - **Immediate Inventory Rebalancing: Internal Stock Transfers** from overstocked hubs (CDMX, Morelia) to high-stockout regions (La Paz, Toluca) to recapture lost revenue without new procurement.
  - **Strategic Replenishment:** Prioritize the restock of the **Top 5 Revenue and Volume items** (Action Figures and Barrel O' Slime).
  - **Marketing & Trend Alignment:** Align Action Figure inventory with **cinematic/anime trends** and introduce **Gachapon/Blind Box** collectibles to drive repeat traffic.
  - **September Recovery Plan:** Launch a "Festival & Heritage" campaign to counteract the forecasted September revenue slump.

Background of Project

Maven Toy is toy store that generates revenue primarily through direct sales of toys to consumers. The company is focusing on sales through direct sales and leveraging data analysis for informed decision-making.

According to Accio.com *"the **global toy and games market** is experiencing **significant growth**..."*

*M*eanwhile Maven Toys revenue has significantly **decrease in 3 months**.



Hypothesis

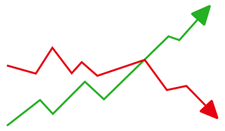
Toy buyers is higher than other category

Sales increase during holidays

Stockouts of the top 5 best-selling toys correlate with decrease sales for those products.

Inventory Aging or Dead Stock in the warehouse

Lower price = Higher volume, but we must looking at Gross Profit Margin

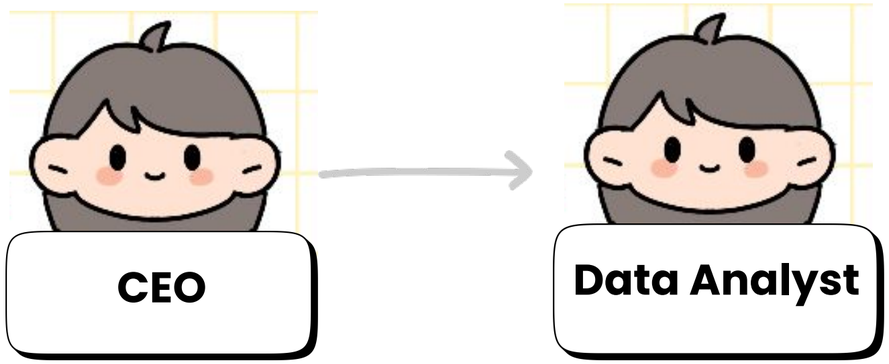


Project Scope & Data Overview

The dataset used in this analysis is the Mexico Toy Sales dataset from Maven Analytics, also utilized in the DEEP program. The dataset is well-suited for retail analytics, as it captures sales transactions, product attributes, inventory levels, and store information, enabling end-to-end business analysis.

The dataset has 4 tabel with more tahn 829k rows in sales and in total has 18 column covering the Maven Toys Store activity in Inventory and Product

Problem Statement



“Which products and price tiers should be prioritized for promotional investment to increase total revenue by 10% within the next 6 months?”

Objective

Identify the key factors products and price tiers that should be prioritized for promotional investment to increase total revenue by 10% within the next 6 months.

Methodology

Business Understanding

Understanding the business question and how to answer question

Data Cleaning

Correcting data type and merging dataset using Python

Data Analysis & EDA

Generate Insight Revenue, Profit, Price Segmentation, Holiday Promo, and Inventory Status (Stockout/Overstock, Weeks of Supply, Turnover)

Data Visualization

Generate Insight and chart top product, Status Product Category in Store Inventory, and Forecast in Tableau and Python

Presentation

Communicate information to stakeholders by giving some recommendation

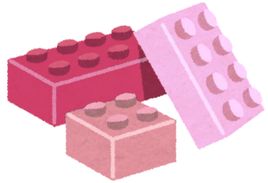
High-priced products contribute 40.5% of revenue, but medium-priced products deliver the strongest profit impact.

- **High Price Tier** (Margin 15.66%, Rev $5.66M)
  - Highest revenue contribution
  - Lower units per sale
- **Medium Price Tier** (Margin 33.7%, Rev $5.23M)
  - Best balance of volume and margin
  - Top profit driver: Colorbud (Electronics) — $834K profit
- **Low Price Tier** (Margin 50.64%, Rev $3.55M)
  - Highest margin %
  - Lower absolute profit due to low price point

Prioritize :

Medium-priced electronics in **promotions** and assortment planning, while using **low-priced items to drive traffic**.

A small number of products drive a disproportionate share of revenue and require focused management.



Revenue $2,4M

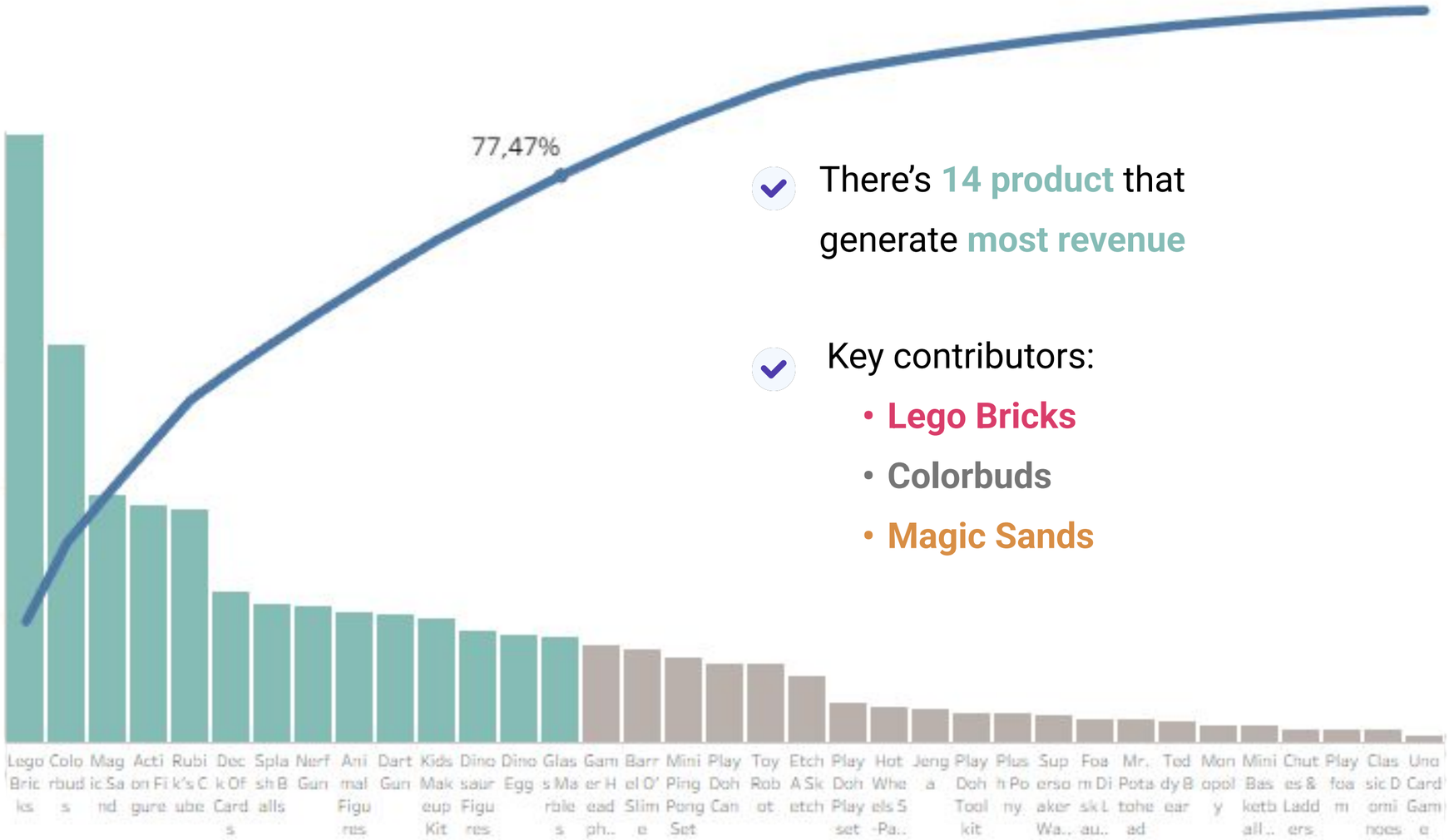
16.4%

Revenue $1,6M

10.8%

Revenue $1M

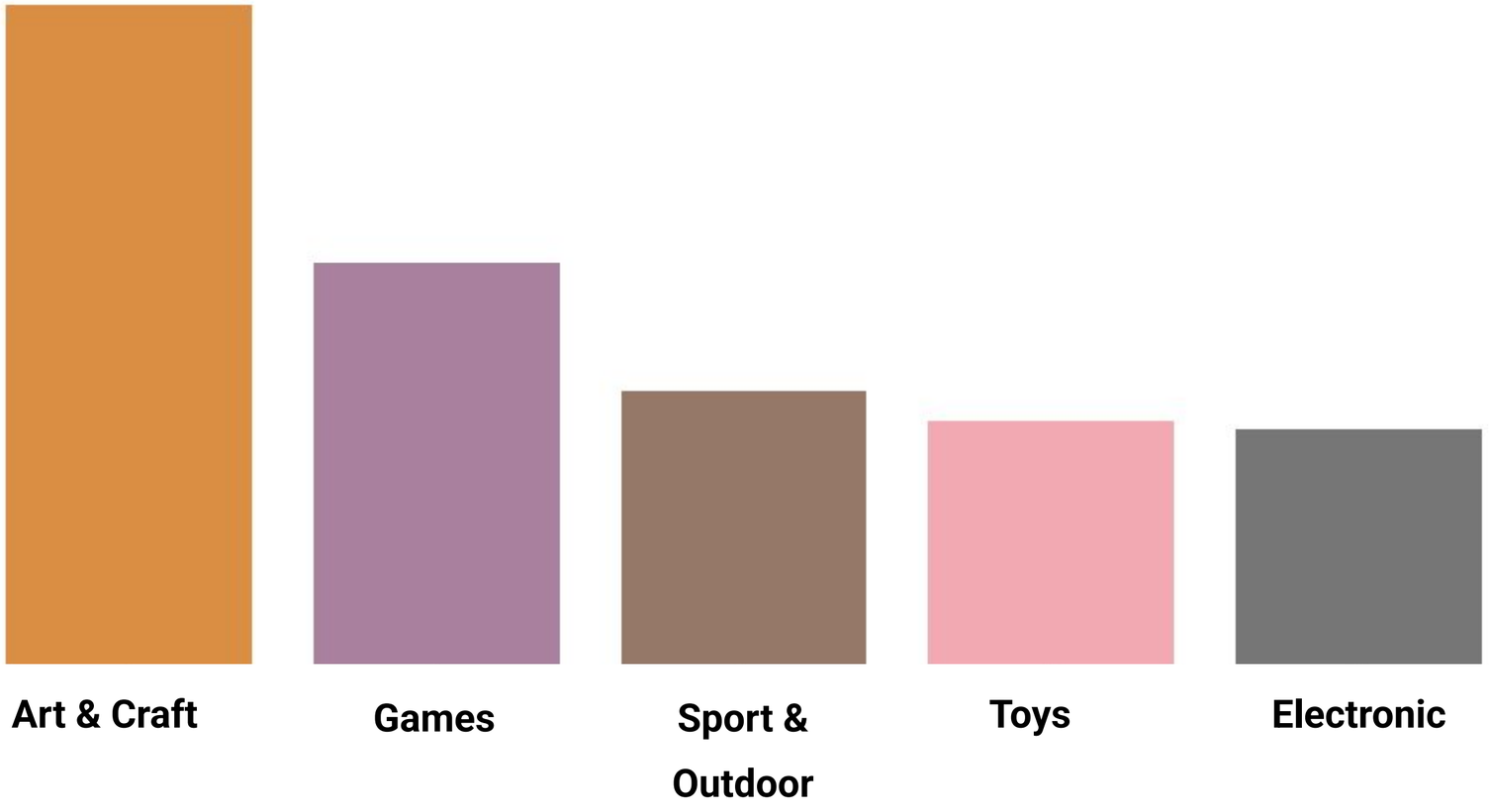
6.7%



- There's **14 product** that generate **most revenue**
- Key contributors:
  - **Lego Bricks**
  - **Colorbuds**
  - **Magic Sands**

Holiday promotions generate a 4% average revenue uplift, with outsized gains in specific categories.

- Holiday (Dec–Jan) Revenue: $2.17M
- Share of Total Revenue: 15.0%
  Category uplift during holidays:
  - Art & Crafts: **+10.7%** (nearly triple store average)
  - Games: **+6.6%**
  - Sports & Outdoors: **+4.5%**
  - Toys: **+4.0%**
  - Electronics: **+3.8%**



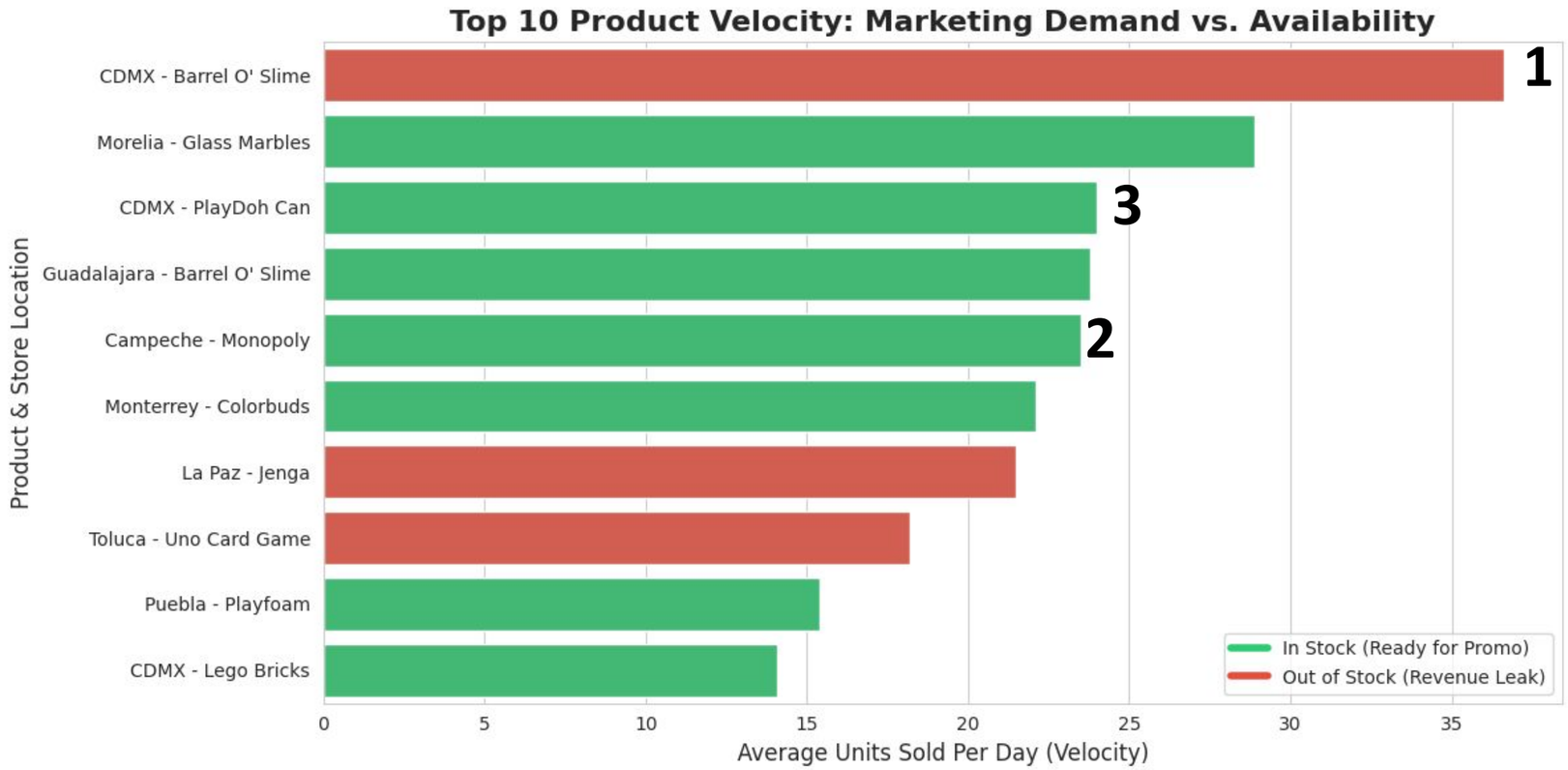
Sales Velocity : Identifying Our High-Traffic Drivers

1. **The Power of Slime: Barrel O' Slime** is our highest-velocity product, moving **36.6 units per day** in Ciudad de México and **23.8** in Guadalajara.

2. **2. The Monopoly Anchor:** In Campeche, **Monopoly** maintains a dominant velocity of **23.5 units per day**, marking it as a critical regional anchor.

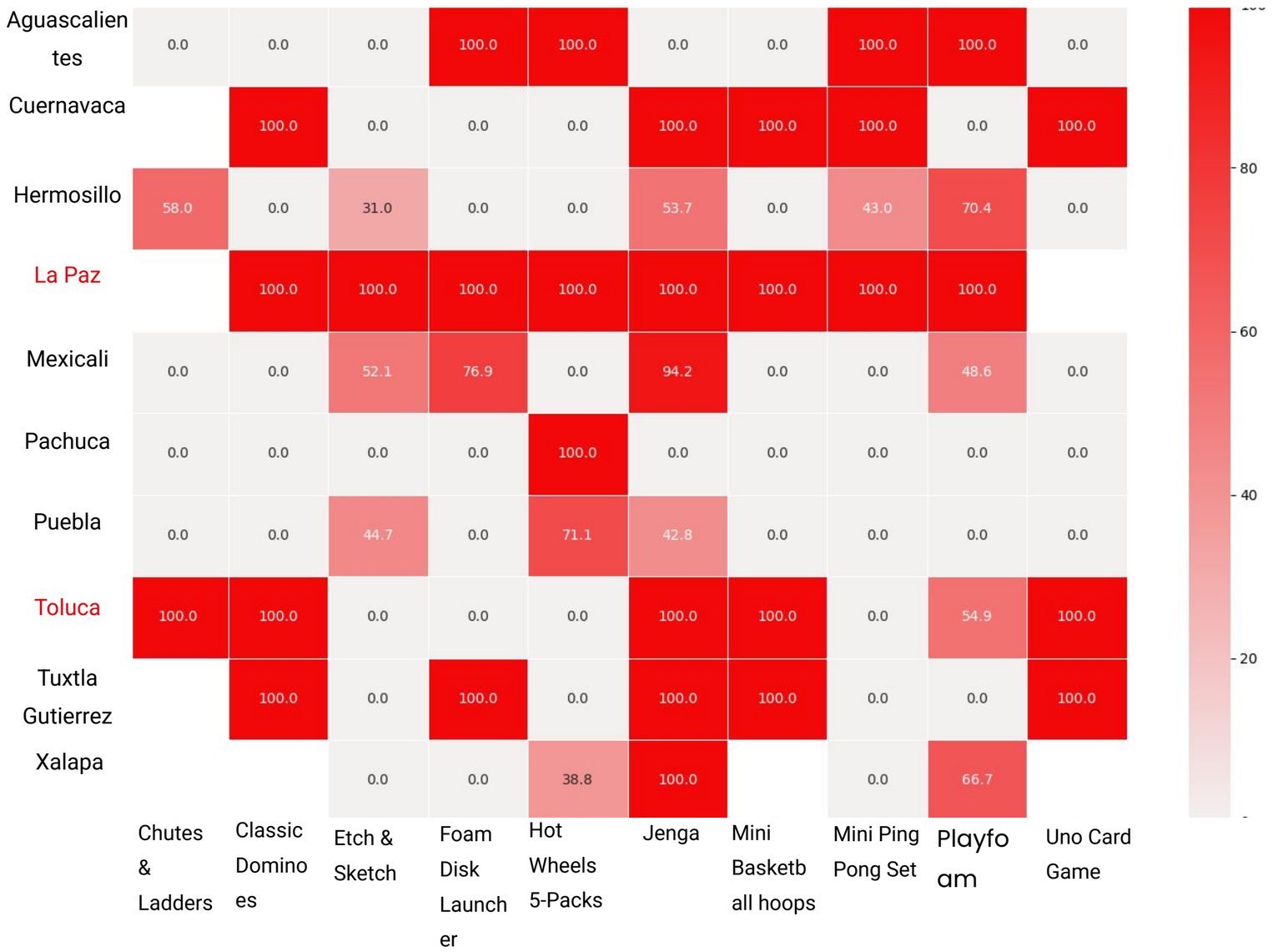
3. **The Revenue Engine:** High-velocity items like **PlayDoh Can (24.0 units/day)** drive the foot traffic that sustains our store ecosystem.

The Bottom Tier: Low-velocity items, such as the **Plush Pony in Oaxaca (4.3 units/day)**, should be evaluated for markdown or replacement to free up shelf space.



Problem: Marketing Demand vs. Zero Availability

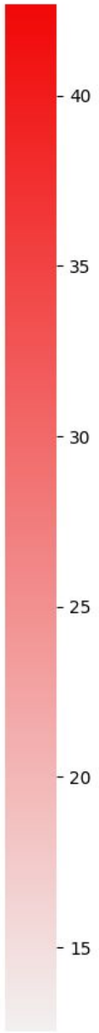
- **The Velocity Gap:** We are driving demand for products with massive velocity, yet our Top-5 revenue **Action Figure** had zero velocity for **354 days** due to stockouts
- **Lost Momentum: Barrel O' Slime**—our #1 velocity leader—suffered a **101-day stockout**, effectively "turning off" our most consistent traffic driver.
- **Regional Mismatch:** We have 100% stockout rates in **La Paz** for high-velocity items, while **Morelia** is overstocked with **Glass Marbles** (28.9 units/day velocity).



Frozen Capital: Inefficient Distribution Is Stalling Growth

- **Excessive Holding Times**: Average inventory sits for 14 weeks
- **"Zombie" Inventory**: Supply holding up to 42 weeks (nearly one year)
- **Action Plan**: Halt new procurement for overstocked categories and initiate Internal Stock Transfers to fill "Red Zone" gaps without additional spend.

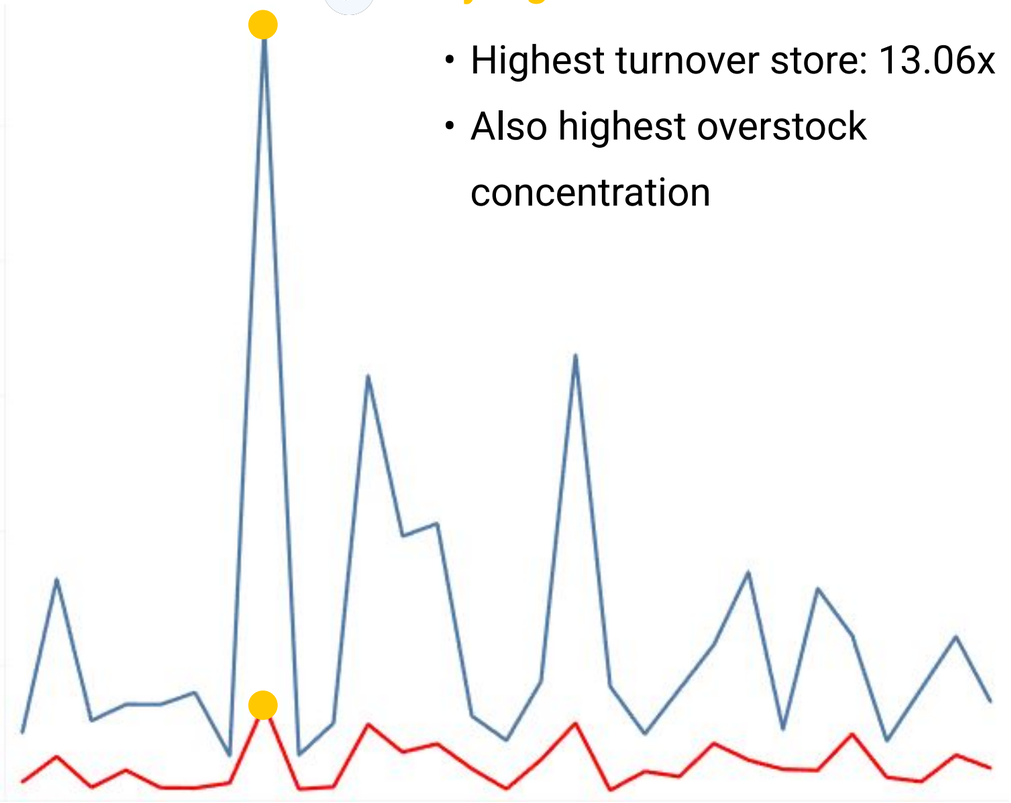
	Art & Craft	Electronic	Games	Sport & Outdoor	Toys
Aguascalientes	35.1	21.1	30.1	12.5	16.1
Campeche	33.8	15.4	27.1	15.2	19.8
Chetumal	24.0	13.9	32.3	17.6	14.5
Chihuahua	20.7	25.4	30.2	26.4	28.8
Chilpancingo	19.9	23.1	42.7	36.9	36.7
Ciudad Victoria	27.0	20.2	33.5	15.5	19.7
Cuernavaca	35.0	24.1	31.2	19.4	16.7
Cuidad de Mexico	29.1	18.6	40.5	18.8	28.5
Culiacan	19.3	16.1	28.6	21.5	21.0
Durango	21.2	13.3	34.4	18.7	24.2
Guadalajara	22.7	25.9	41.0	15.8	27.6
Guanajuato	20.9	14.7	31.1	23.3	25.3
Hermosillo	27.8	21.3	31.9	31.3	30.6
La Paz	27.0	13.9	24.2	33.7	27.0
Merida	33.7	20.8	30.8	20.8	24.3
Mexicali	26.7	17.2	34.4	17.3	34.8
Monterrey	27.9	15.2	29.6	21.5	19.7
Morelia	41.6	16.5	27.9	21.3	18.1
Oaxaca	23.4		21.6	27.4	24.0
Pachuca	30.2	16.4	23.9	22.1	23.5
Puebla	18.5	15.0	26.3	17.5	31.6
Saltillo	21.5	34.3	34.0	22.9	20.3
San Luis Potosi	18.1	20.0	20.7	31.8	28.2
Santiago	32.8	12.7	38.5	15.7	19.1
Toluca	21.7	16.4	26.6	19.8	25.1
Tuxtla Gutierrez	24.5	21.9	20.3	21.2	22.0
Villahermosa	35.7	15.4	29.2	19.6	33.7
Xalapa	23.6	14.1	35.5	30.3	23.2
Zacatecas	30.0	16.3	17.4	15.0	28.4



The Cuidad de Mexico Paradox: High Velocity vs. Overstock Risk

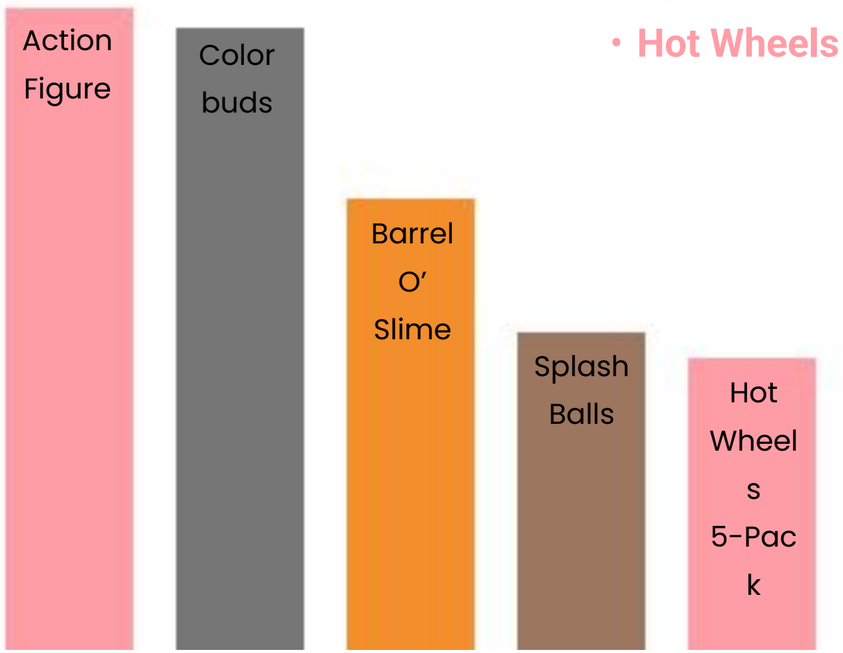
City Signal : Cuidad de Mexico

- Highest turnover store: 13.06x
- Also highest overstock concentration

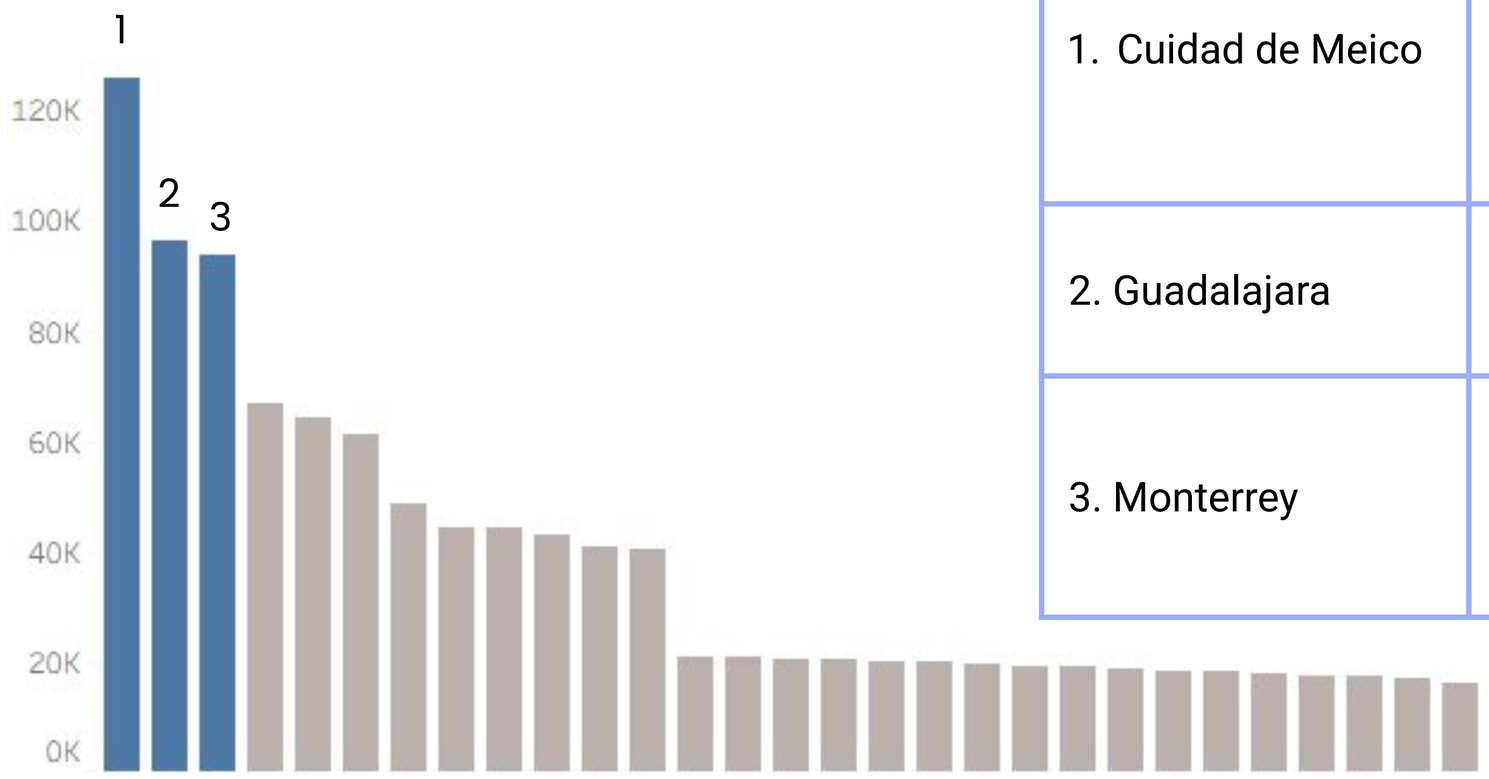


5 products need attention:

- **Action Figures** (25.9%)
- **Colorbuds** (25.4%)
- **Barrel O' Slime** (20.9%)
- **Splash Balls** (17.5%)
- **Hot Wheels 5 Pack** (16.8%)



Revenue & Units Are Concentrated in 3 Cities

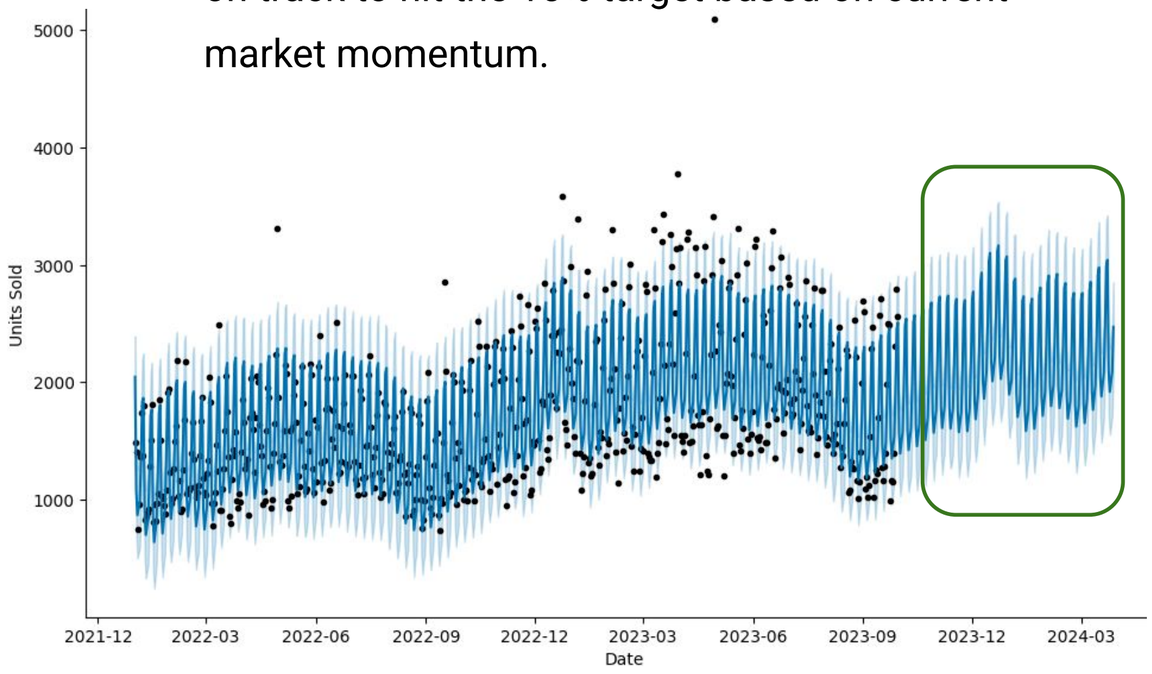


City	Units Sold	Winning Product
1. Cuidad de Meico	125k	• Barrel o Slime, • Colorbuds, • Deck of Card, • Playdoh Can
2. Guadalajara	96k	• Colorbuds • Playdoh Can
3. Monterrey	93k	• Colorbud, • Playdoh Can, • Splash Balls

Growth Forecast: Navigating the September Slump

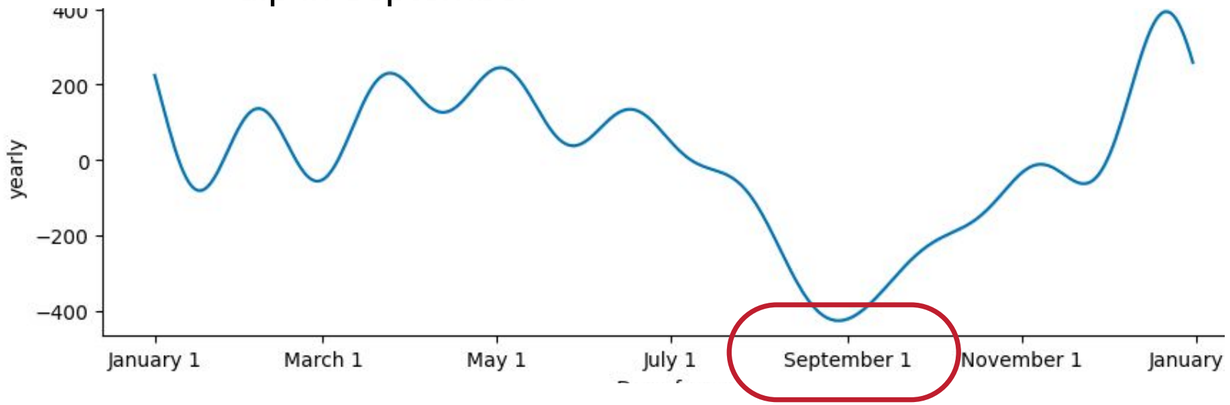
Positive Trajectory

Forecast models confirm that Maven Toys is on track to hit the 10% target based on current market momentum.



The Hidden Risk

Python Prophet identifies a significant forecasted dip in September



Strategic Pivot

Launch a "September Recovery Promo" to turn this forecasted loss into a realized gain.

Recommendation

Quick Wins

Reallocate inventory

- Move Barrel O' Slime, Playdoh, Action Figure to top 3 cities

Minimum stock rule

- Enforce safety stock on Top-10 volume SKUs

September readiness

- Pre-build inventory buffer by August

Major Projects

Inventory Turnover Governance

- Monthly turnover review by product & city

Promo + Inventory Sync

- No promotion without stock coverage

City-level replenishment logic

- Mexico City treated as separate supply cluster

Innovation Plan

Trend-Alignment for Action Figures

- Aligning inventory with current Anime and Cinematic trends

The "Collector" Strategy (Gachapon & Blind Boxes)

- Exclusive IP Collaborations and increase customer lifetime value.

The "September Slump" Counter-Strategy

- Reposition 'Art & Craft' and 'Music' sectors

Key Takeaways

- **Recapture the $3.7M Gap**: Restocking the Top-5 revenue and volume drivers (Action Figures and Barrel O' Slime)
- **Solve the Regional Crisis**: stock transfers from overstocked locations.
- **Optimize Capital Flow**: Reduce the average 14-week inventory holding time by liquidating stock in categories with >18 weeks of supply
- **Market-Led Innovation**: Counteract the forecasted September dip by aligning Action Figure inventory with cinematic trends and introducing Gachapon/Blind Box collectibles.

Summaries

- **The Stockout Reality**: 25% of the catalog is at immediate risk, with a Top-5 revenue Action Figure missing from shelves for 354 days.
- **The Overstock Reality**: Maven Toys carrying up to 2 years of supply for low. This imbalance represents a distribution failure, not a demand failure.
- **Macro Trend**: Both forecast models confirm a positive trajectory toward the 10% growth target.
- **The September Risk**: Advanced modeling identifies a seasonal slump in September.
- **Winning Segments**: Colorbuds remains our profit leader, while Lego Bricks drives premium revenue through price

OBIPR Frameworks

Observation	Business Impat	Isolation	Prioritization	Recommendation
• Revenue forecast shows +10% growth potential • Repeated stockouts in top-selling products • High turnover and high overstock coexist in key cities • September shows unexpected unit decline	• Lost revenue from unavailable bestsellers • Inefficient capital tied in slow inventory • Promotion effectiveness weakened • Risk of missing growth target despite demand	Replenishment not aligned with: • Product velocity • City-level demand Inventory policy treats all stores equally Promotions planned without stock validation	• High- Inventory availability : Direct revenue impact • High- Top 4 SKUs: Concentrated demand • Medium- Forecast integration: Risk prevention • Low- New SKU expansion: Not growth blocker	Immediate! • Fix stock availability for Top-10 volume SKUs Mid-Term • Implement turnover-based replenishment • Align promo calendar with inventory readiness Strategic • Use forecast as planning input, not reporting output

Dashboard

Disclaimer

- This analysis is based on publicly available Maven Analytics dataset and does not reflect real-world financial or business insight.
- Maven Toys Store is a fictional entity and the result presented here are purely for educational purposes.
- The queries and insight generated in this assignment should be personalized to my own analysis results.

Appendix

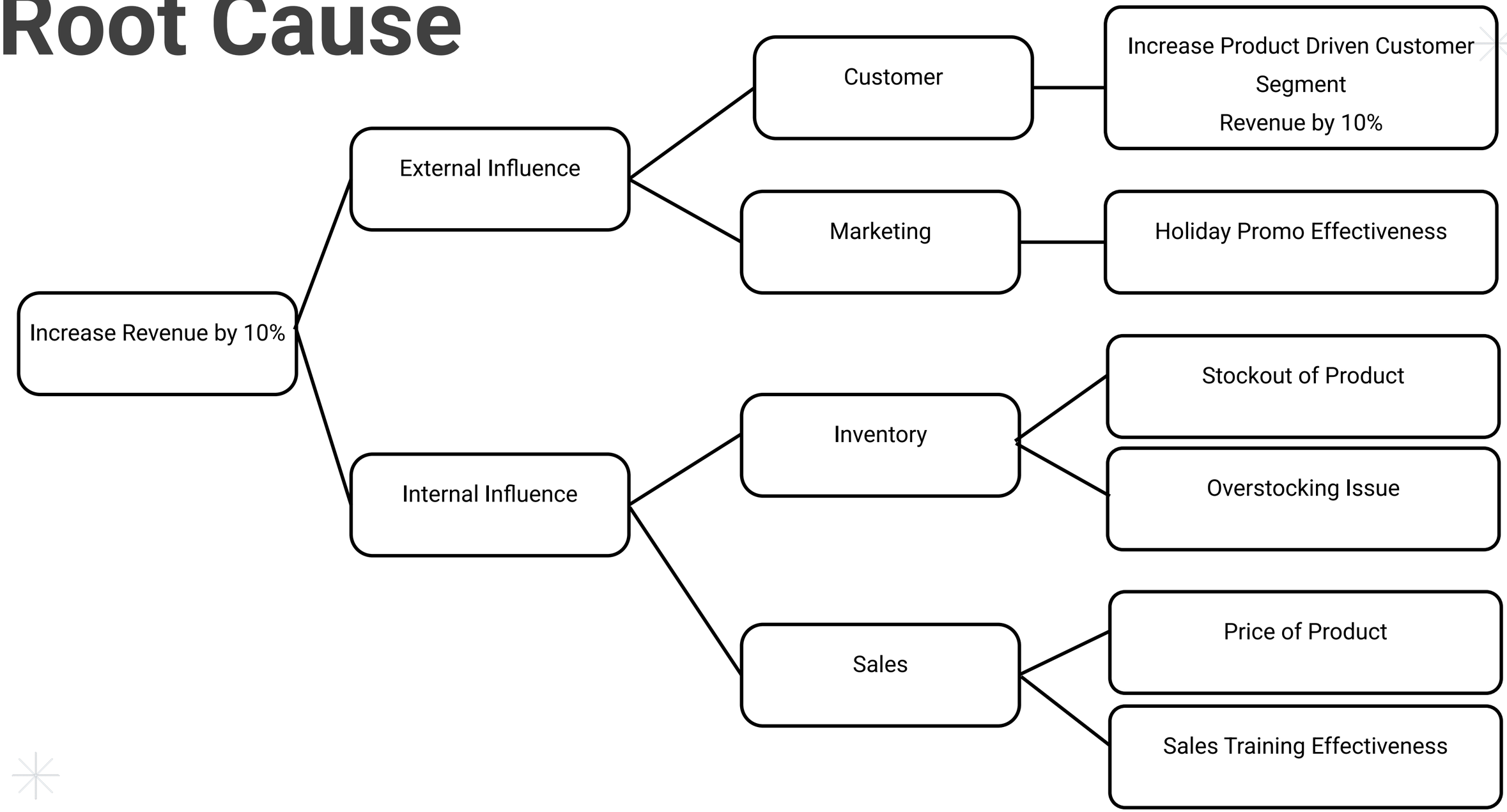
Calculation : Python

Calculation : Tableau

Stakeholder

Decider	Head of Marketing
Accountable	Head of Analyst
Responsible	Data Analyst Team
Consultant	Head of Finance, Head of Logistics
Inform	Marketing Teams, Logistic Teams,

Root Cause


Increase Revenue by 10%
External Influence
Internal Influence
Customer
Marketing
Inventory
Sales
Increase Product Driven Customer Segment Revenue by 10%
Holiday Promo Effectiveness
Stockout of Product
Overstocking Issue
Price of Product
Sales Training Effectiveness

Weeks of Supply

If WoS > 12, over 3 months of stock (Overstock)
If WoS < 2, at risk of running out (Urgent Restock)

- At least 25% of products have less than 5 weeks of supply. Given shipping times and lead times, many of these are likely hitting zero (the min of 0.00 proves this).
- The average product sits for 14 weeks (over 3 months!). Even worse, some products have 104 weeks of supply—that is 2 years of stock sitting in a warehouse.

count	1590.000000
mean	14.093934
25%	4.884700
Median	10.168395
75%	18.181818
Max	104.934354

Sales Velocity

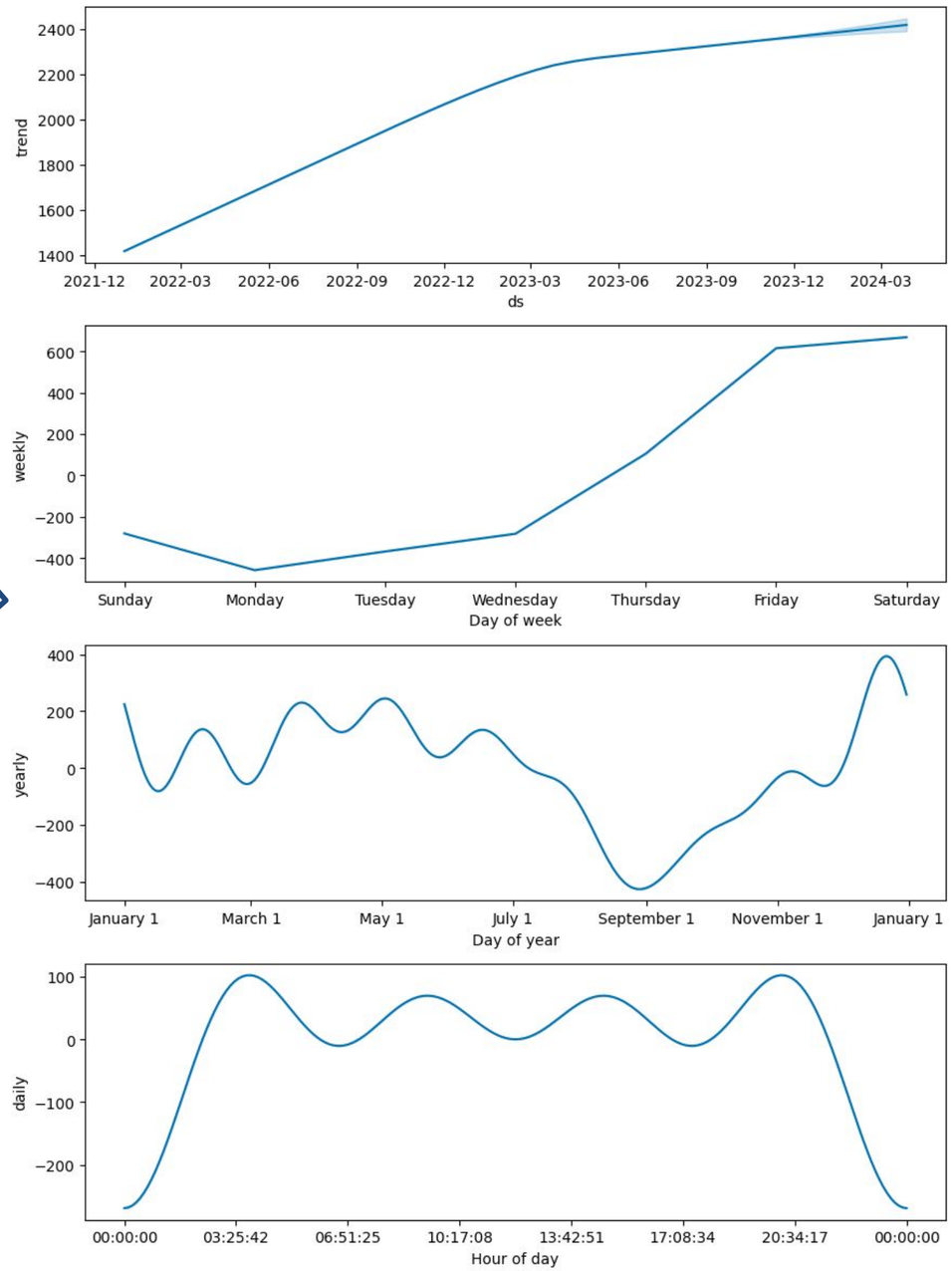
```
daily_sales = df_sips.groupby(['Store_City', 'Product_Name',
'Date'])['Units'].sum().reset_index()

# Filter out days with 0 sales (This removes the Stockout noise)
# We only want to measure velocity when the product was actually 'On Shelf'
active_sales = daily_sales[daily_sales['Units'] > 0]

# Calculate Average Units per Day (Sales Velocity)
velocity_df = active_sales.groupby(['Store_City',
'Product_Name'])['Units'].mean().reset_index()
velocity_df.rename(columns={'Units': 'Sales_Velocity_Per_Day'}, inplace=True)

# 4. Merge back to your main data to see the gap
# Velocity vs. Total Opportunity
print(velocity_df.sort_values(by='Sales_Velocity_Per_Day', ascending=False))
```

	Store_City	Product_Name	Sales_Velocity_Per_Day
235	Cuidad de Mexico	Barrel O' Slime	36.599388
588	Morelia	Glass Marbles	28.926554
257	Cuidad de Mexico	PlayDoh Can	24.006993
335	Guadalajara	Barrel O' Slime	23.757576
56	Campeche	Monopoly	23.521739
...	...	...	...
876	Tuxtla Gutierrez	Uno Card Game	5.333333
666	Pachuca	Playfoam	5.333333
148	Chilpancingo	Etch A Sketch	5.333333
502	Merida	Teddy Bear	5.250000
633	Oaxaca	Plush Pony	4.333333

972 rows × 3 columns

Forecast

Forecast by Unit in Python

```
df_prophet = df_sips.groupby('Date')['Units'].sum().reset_index()
df_prophet.columns = ['ds', 'y']
m = Prophet(yearly_seasonality=True,
daily_seasonality=True).fit(df_prophet)
future = m.make_future_dataframe(periods=180)
forecast = m.predict(future)
```

Forecast by Revenue in Tableau